

Name: _____

10.1 Tangents.

1. A line meets a circle at exactly two points. What is it called? What is the segment between those points called?
2. A radius of 10 meets a 24-unit segment at the circle, whose far end is 26 from the center. Tangent? Show the check.
3. A tangent segment is 15 and the secant through the center has external piece 9. Find r : $15^2 + r^2 = (r + 9)^2$.
4. Tangent segments from one point are $6x - 5$ and $4x + 9$. Find x and the length.

10.2 Arcs.

5. AB is a diameter and $m\widehat{AC} = 57^\circ$. Find $m\widehat{CB}$.
6. A minor arc measures 108° . Find its major arc.
7. A category is 40% of a circle graph. Find the central angle.

10.3 Chords.

8. Chords $PQ \cong RS$ in one circle, $m\widehat{PQ} = 77^\circ$. Find $m\widehat{RS}$.
9. A diameter is perpendicular to a 34-unit chord. Find each piece.
10. Radius 26; a chord is 10 from the center. Find the chord.

10.4 Inscribed angles.

- 11.** An inscribed angle intercepts a 146° arc. Find it.
- 12.** A triangle in a semicircle has acute angles $(5x + 1)^\circ$ and $(2x + 5)^\circ$. Find x and both angles.
- 13.** An inscribed quadrilateral has opposite angles $(3y + 20)^\circ$ and $(y + 40)^\circ$. Find y and both angles.

10.5 Angle relationships.

- 14.** A tangent-chord angle is 59° . Find its arc.
- 15.** Two chords cross; vertical arcs 101° and 87° . Find the angle.
- 16.** Two secants: far arc 144° , near arc 52° . Find the angle.

10.6 Segment products.

- 17.** Chord pieces 8 and 6 meet pieces x and 12. Find x .
- 18.** Secants: external 7 far 9; external 8 far x . Find x .
- 19.** A tangent t meets a secant with external 5, far part 20. Find t (exact).

10.7 Coordinate plane.

- 20.** Write the equation of the circle with center $(-2, 7)$ and radius 9. Then name one point on it.